Spatial–Spectral Enhancement and Fusion Network for Hyperspectral Image Classification With Few Labeled Samples

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***Abstract*— Deep learning has shown great potential in hyperspectral image (HSI) classification. However, training these models usually requires a large amount of labeled data. Since the collection of pixel-level annotations for HSIs is laborious and time-consuming, developing algorithms that can yield good performance in a small sample size situation is of great significance. Therefore, many research works focus on building a deep learning model for HSI classification with few labeled samples. However, prevalent solutions are unsatisfactory in feature discrimination and model overfitting, which greatly limits their performance. To remedy these drawbacks, we propose a novel spatial–spectral enhancement and fusion network for hyperspectral image classification with few labeled samples, named SSEFN. Specifically, we design a spatial–spectral enhancement strategy (SSES) to boost the feature discrimination from spatial and spectral perspectives, which enables the model to learn more easily with fewer samples. In addition, we propose an adaptive decision fusion (ADF) module to fuse the decisions of all enhanced features. Since each decision prediction may have a different trend of overfitting, combining multiple predictions alleviates the overfitting. Extensive experiments are conducted on four diverse hyperspectral image datasets. The results show that our method significantly outperforms the state-of-the-art approaches on all datasets and demonstrates the effectiveness and superiority of the proposed model for hyperspectral image classification with few labeled samples. Codes are available at https://github.com/liushuang963/SSEFN.**

***Index Terms*— Decision fusion, few labeled samples, hyperspectral image (HSI) classification, spatial–spectral feature enhancement.**

I. Introduction

With the advancement of Earth observation technology, hyperspectral image (HSI) can be widely acquired [1]. Unlike traditional vision systems that capture images only containing RGB channels, HSIs encompass a much broader spectral range, covering the visible spectrum, near-infrared, mid-infrared, and far-infrared regions, with dozens or even hundreds of continuous wavebands [2], [3], [4]. Benefiting from the abundant and detailed spectral information, HSIs provide an opportunity to identify ground objects that are difficult to distinguish in natural RGB images [5]. This makes more accurate observations of the Earth possible [3]. Due to these advantages, HSIs are widely applied in various applications [6], including urban mapping, resource exploration, and environmental monitoring. As the fundamental task of these applications, HSI classification is to predict the class label of each pixel in the image [7].

In early research on HSI classification, a large number of machine learning methods were developed, such as the support vector machine (SVM) [8], random forest (RF) [9], multinomial logistic regression (MLR) [10], and the feature selection or feature extraction method [11], [12]. However, these are shallow models that rely on expert prior knowledge [3], making it difficult to strike a balance between discriminability and robustness [13]. To address this drawback, deep learning [14] has been introduced to hyperspectral image classification, and it achieves good results, benefiting from strong hierarchical feature extraction capabilities. The current mainstream hyperspectral image classification methods based on deep learning can be broadly divided into a convolutional neural network (CNN-based) [15], [16], [17], [18], Transformer-based [19], [20], diffusion-based [21], [22], Mamba-based [23], [24], and so on [25] and [26]. CNN-based models [27] have strong spatial pattern extraction capabilities, while Transformer-based [28] models have strong global modeling capabilities. The iterative nature of diffusion models facilitates the modeling of the spatial–spectral distribution of high-dimensional data [21]. Mamba can model the global relationship of sequences with linear complexity, improving the ability to extract global features [29], [30]. However, the above methods generally require a large amount of labeled data, attributable to the requirement of learning and optimizing thousands or even millions of parameters [31]. Unfortunately, the collection of pixel-level annotations for HSIs is both laborious and time-consuming [32]. Therefore, it is of significant importance to develop algorithms capable of delivering robust performance under small sample size conditions [33]. Therefore, it is of significant importance to develop algorithms capable of delivering robust performance under small sample size conditions [33].

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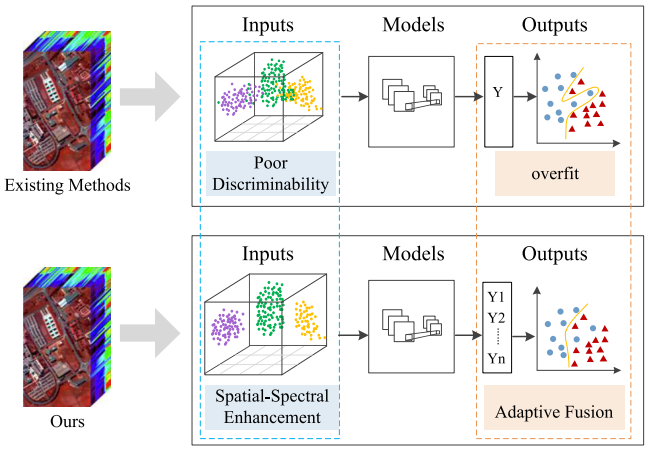


Fig. 1. Existing methods suffer from the problems of poor feature discrimination and easy overfitting in the case of few samples. In contrast, we propose an SSES to improve feature discrimination and an ADF module to reduce the overfitting.

In recent years, some hyperspectral few-sample work have been proposed. These methods can be mainly divided into the following four categories: data enhancement [18], [34], [35], transfer learning [36], [37], [38], few-shot learning (FSL) [39], [40], and semi-supervised learning [41], [42], [43]. Data enhancement utilizes related strategies to convert training samples into many different samples, allowing the model to gain greater generalization capabilities [34]. Transfer learning attempts to transfer knowledge learned from the source domain with abundant labeled samples to the target domain with limited labeled samples [44]. FSL is different from most other deep learning methods in that it aims to study the differences between samples rather than directly learning what the samples are [33]. Semi-supervised learning aims to improve classification accuracy by exploiting an additional set of unlabeled samples [41]. However, these works either use raw data for training or adopt enhancement strategies for natural images, such as Gaussian noise enhancement strategies [35], [45]. These strategies do not fully consider the characteristics of the hyperspectral image, such as the continuity of spectral and multiscale characteristics of ground objects, resulting in low discriminability of the enhanced data. In addition, when these algorithms have few samples, these models are prone to overfitting (see Fig. 1).

To this end, we propose a novel spatial–spectral enhancement and fusion network for HSI classification with few labeled samples, termed SSEFN, which can simultaneously extract discriminative features and reduce overfitting. Specifically, we first design the spatial–spectral enhancement strategy (SSES), including the spatial enhancement strategy (SpaES) and the spectral enhancement strategy (SpeES), to boost the feature discriminability from spatial and spectral perspectives. The SpaES extracts multiple scale patches to help the identification of multiscale ground objects. The SpeES generates multiscale spectral difference information, which enhances the discriminability of spectral features. The enhanced spatial and spectral features are then fed into subnets to obtain the corresponding predictions. Finally, these predictions are fused by the proposed adaptive decision fusion (ADF) module to obtain the final prediction. Since each prediction may have a different overfit trend, integrating multiple predictions at the decision level can mitigate the overfitting problem when there are few labeled samples. We conducted extensive experiments on diverse real-world datasets, and the results demonstrate that our method significantly outperforms the state-of-the-art approaches. In summary, the main contributions of this article are as follows.

1) We propose a novel spatial–spectral enhancement and fusion network for hyperspectral image classification with limited training samples, named SSEFN, which has strong feature extraction capability and is robust for overfitting on limited samples.
2) We design two information enhancement strategies, that is, the SpaES and the SpeES, to provide more discriminative information for HSI classification. Therefore, the model can learn more easily with fewer samples.
3) We develop an ADF module to fuse the decision of all enhancement information. Since each prediction may have a different overfitting trend, the fusion of multiple predictions alleviates the overfitting when the training samples are few.

II. Related Work

A. Hyperspectral Image Classification

HSI classification methods can be divided into two main categories: traditional machine learning [8], [10] and deep learning [46], [47]. Most of the former are shallow models that rely on expert prior knowledge [3], mainly including SVMs [8], RFs [48], and MLR [10]. Peng et al. [49] provided a theoretical analysis of joint sparse representation and employed a maximum likelihood estimation strategy to effectively handle inhomogeneous neighboring pixels or outliers. In [9], the RF is used to classify the spatial–spectral features of HSIs through two ensemble strategies. Peng et al. [50] proposed a self-paced JSR model that can adaptively select good pixels and effectively handle the noisy neighboring pixels or spectral bands. The latter does not rely on expert knowledge and can automatically learn features from data that discriminate the classification task [6]. Therefore, more and more research has adopted deep learning networks to solve HSI classification problems. The CNN can automatically capture the hierarchy characteristics from shallow to deep and has a strong spatial pattern extraction capability [51]. Hence, a series of CNN-based HSI classification methods [46], [47], [51] have been proposed. Huang et al. [52] proposed a novel two-branch attention adversarial domain adaptation (TAADA) network for hyperspectral image classification, which embedded domain adaptation into the process of adversarial learning to learn discriminative and domain-invariant features and effectively solved the distribution discrepancy problem in the cross-scene hyperspectral image classification. Nevertheless, these CNN-based methods cannot model long-range dependencies due to the inherent locality of the CNN, which hinders

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the feature extraction ability of the models [53]. In contrast, the Transformer can directly model relations between all tokens by the self-attention mechanism and has strong long-distance dependency modeling capabilities. For example, SSFTT [54] employs Gaussian distribution weighted tokenization to transform shallow spectral–spatial features into tokenized semantic features. Chen et al. [28] proposed a framework that can achieve adaptive spectral–spatial information fusion using the Spectral Query Spatial paradigm, reducing the introduction of irrelevant information and effectively improving classification performance. Besides, SpectralDiff [21] can facilitate cross-sample perception via reconstruction distribution, thus improving classification performance. MTMSD [22] explores multitimestep multistage diffusion features for HSI classification. In addition, MambaHSI [29] reveals the great potential of Mamba to be the next-generation backbone for HSI models. HyperMamba [30] effectively captures spatial and spectral information while maintaining the computational efficiency of the Mamba architecture.

However, the aforementioned approaches generally require a large dataset and adequate labeled data to train models with good generalizability [41]. Unfortunately, the collection of pixel-level annotations for HSIs is both laborious and time-consuming [32]. To this end, we propose a novel spatial–spectral enhancement and fusion network for hyperspectral image classification with few labeled samples, which has strong feature extraction capability and can achieve superior performance on few labeled samples.

B. HSI Classification With Few Labeled Samples

As mentioned above, the problem of limited training samples is one of the main obstacles affecting the accuracy of HSI classification. Researchers have proposed many algorithms to improve the classification results of a few training samples. These techniques can be roughly divided into the following four types: data enhancement [34], [35], [55], FSL [56], [57], transfer learning [44], [58], and semi-supervised learning [41], [59]. Data enhancement aims to enrich sample data and improve the generalizability of the training model [34], [55]. For example, Gong et al. [34] proposed a data augmentation strategy called quad-patch augmentation to enrich the training set and introduced a dual-path aggregated attention module to focus on key cues. Chen et al. [35] added Gaussian noise to the original data to generate augment data for training. FSL aims to achieve robust representations without relying on large-scale datasets [56]. Tang et al. [57] introduced the prototype network to obtain prototypes of all classes and then classify samples by computing distances to the prototype of each class. Liu et al. [60] proposed a refined prototypical contrastive learning network for FSL (RPCL-FSL), which incorporated supervised contrastive learning (SCL) and FSL into an end-to-end network for the first time and designed triple-constraint-based prototypical learning modules to stabilize the prototypes. Transfer learning utilizes the knowledge implied in the source domain(s) with abundant labeled instances to improve the performance of the learned model in the target domain(s) with limited labeled samples [44]. Zhang et al. [58] pretrained the 3D-CNN on the source HSI dataset and then fine-tuned the model on the target HSI dataset with limited labeled samples. Semi-supervised learning can exploit an additional set of unlabeled samples to enhance the model performance [41]. For instance, the semi-supervised methods, that is, self-training [61], co-training [62], and generative model [63] transfer knowledge from unlabeled samples to increase the robustness of the model for limited labeled samples.

However, these works either use raw data for training or adopt enhancement strategies for natural images, such as Gaussian noise enhancement strategies [35], [45]. These strategies do not fully consider the characteristics of the hyperspectral image, such as the continuity of spectral and multiscale characteristics of ground objects, resulting in low discriminability of the enhanced data. In addition, when these algorithms have few samples, these models are prone to overfitting. In contrast, in terms of spatial dimension, we proposed a SpaES to enhance spatial multiscale feature extraction; in terms of spectral dimension, we designed a SpeES, introduced spectral continuity before estimating multiscale spectral gradients, and enhanced spectral feature extraction. Hence, the model can learn more easily with fewer samples. Furthermore, we propose an adaptive decision-fusion module to fuse all the corresponding predictions of each enhanced spatial and spectral feature. Since each prediction may have a different trend of overfitting, the fusion of multiple predictions can mitigate the overfitting when training samples are few.

III. Proposed Method

A. Motivation and Overall Idea

Training deep learning models usually requires a large amount of labeled data. Unfortunately, the collection of pixel-level annotations for HSIs is laborious and time-consuming. This compels us to develop an HSI classification method, which can achieve superior performance on a few labeled samples. There are two main challenges for HSI classification with limited labeled samples: 1) how to reduce the difficulty of hyperspectral classification and therefore reduce the data requirements of the model; and 2) how to alleviate the overfitting problem of the model when there are few labeled samples. Considering that deep learning is to essentially learn a mapping function from input features to labels, we can observe that if the model input features are more discriminative to the target task, the learning difficulty of the model will be lower and the amount of data will be smaller. Based on the assumption, we propose an SSES to introduce multiscale priors of ground objects and differential information of spectra, which enhances the discriminability of input features (solving challenge 1). In addition, Ying et al. [64] pointed out that the degree of overfitting is affected by factors such as noise in the data, training sample size, model complexity, and so on [65]. Srivastava et al. [66] showed that multiple predictions often have different overfitting trends, and integrating multiple predictions from different models can alleviate the overfitting. Inspired by [64], [65], and [66], we utilize diverse spatial–spectral enhancement features to obtain multiple predictions and then perform ADF to reduce overfitting.

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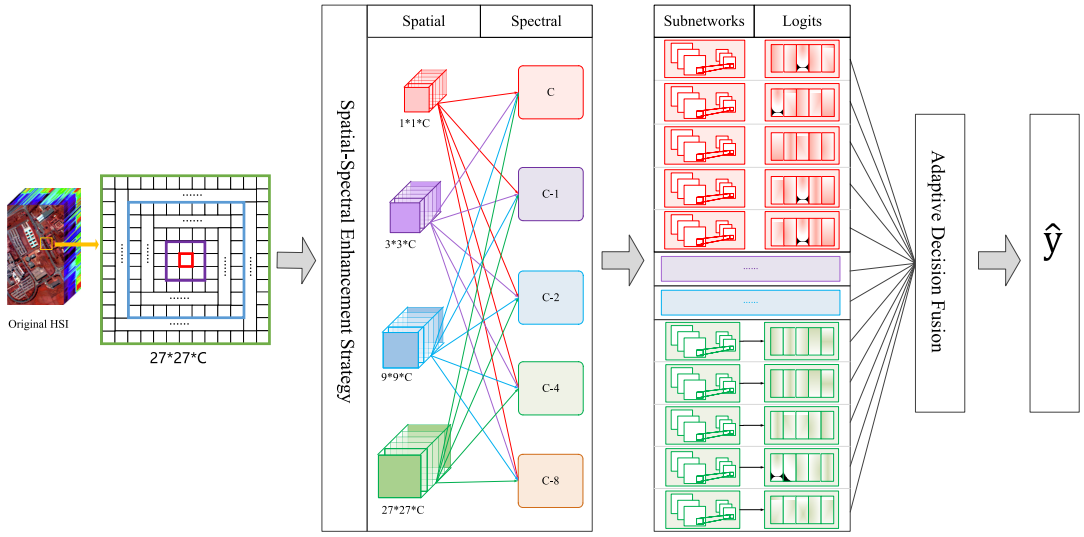


Fig. 2. Overview of the proposed SSEFN framework. A 27 × 27 patch is extracted from the original HSI, and then multiscale spatial enhancement is performed to obtain four spatial information enhancement results. The multiscale contains 1 × 1, 3 × 3, 9 × 9, and 27 × 27. Then, the spectral difference is performed five times for each spatial information enhancement result, and 20 spatial–spectral enhancement results are obtained. In addition, these enhancement results are passed through the subnetwork model to obtain 20 logits. Finally, the 20 logits use the ADF strategy to acquire a reliable prediction result.

B. Overview

An overview of the proposed SSEFN is shown in Fig. 2. The framework contains three components: SSES, subnetworks, and ADF module. The SSES generates more discriminative features for HSI classification. Therefore, the model can learn more easily with fewer samples. Subnetworks are adopted to obtain the corresponding predictions of all enhancement features. The ADF module integrates all predictions with different overfitting trends to obtain the final prediction, thus alleviating overfitting when training samples are few.

C. Spatial–Spectral Enhancement Strategy

The deep model essentially learns a mapping from input x to label y on the training set, and the mapping must be able to adapt to various test data. This means that the model is related to two factors: one is the difficulty of mapping input x to output y, and the other is the learning ability of the network. The more discriminative the sample features are, the easier it is for the model to recognize the differences between different categories. Besides, both spatial and spectral are vital for HSI classification [3]. Therefore, we enhance feature discriminability from both spatial and spectral perspectives, thus improving the model's generalization ability with few labeled samples.

1) Spatial Enhancement Strategy: The size of ground objects in the real world varies greatly, and ground objects of different scales require features of different scales to be recognized [51]. Hence, we propose a SpaES in Fig. 3 to extract multiscale spatial features and improve the discriminability of spatial features. We define (i, j) as the index of a pixel classified by the model. Then, a 27×27 patch, centered on the classified pixel (i, j), is extracted. The process can be expressed as follows:

$$\mathbf{P}_{i,j} = \mathbf{I}[i-13 : i+14, j-13 : j+14, :] \tag{1}$$

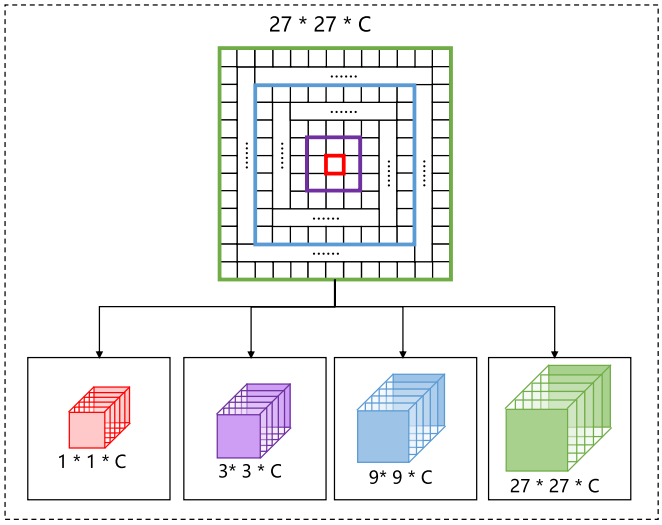


Fig. 3. Architecture of the proposed SpaES. There are four spatial scales: 1 × 1, 3 × 3, 9 × 9, and 27 × 27, which can fully capture the multiscale information of ground objects.

where $\mathbf{I}$ and $\mathbf{P}_{i,j}$ denote the original hyperspectral image and the patch extracted from the original HSI with the center of (i, j), respectively.

Next, we extracted four different scale patches from the original 27×27 patch as follows:

$$\mathbf{P}_{\text{spa}}^{m} = \mathbf{P}_{i,j}\left[i-\frac{m-1}{2} : i+\frac{m+1}{2}, j-\frac{m-1}{2} : j+\frac{m+1}{2}, :\right] \tag{2}$$

where m is the scale of SpaES choosing from $\{1, 3, 9, 27\}$. $\mathbf{P}_{\text{spa}}^{m}$ is the result of spatial enhancement at the mth scale.

2) Spectral Enhancement Strategy: Different objects may have the same spectrum and the same object may appear with different spectra under different background [67]. This makes HSI classification more difficult, so it is necessary to enhance

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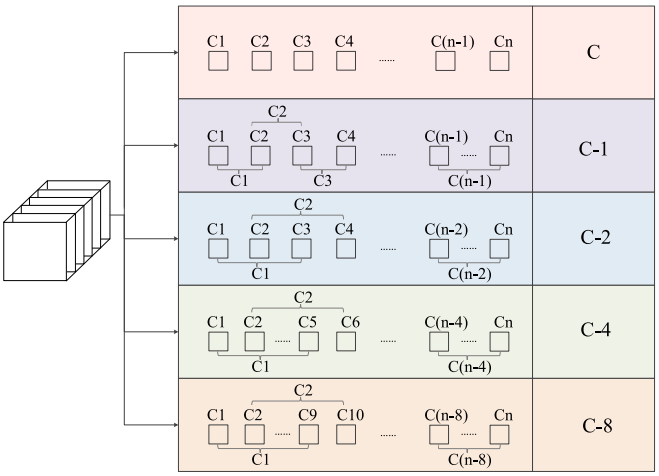


Fig. 4. Architecture of the proposed SpeES. Based on the spatial multiscale method, we propose the spectral difference method for rich spectral information of hyperspectral images. C means the original HSI feature. The difference method included $(C-1)$, $(C-2)$, $(C-4)$, and $(C-8)$, which represent difference every 1 channel, 2 channels, 4 channels, and 8 channels, respectively.

the spectral features of HSIs. Inspired by [68] and [69], the difference in spectra can improve the discriminability of spectral features and the accuracy of HSI classification. The different ground objects have different difference differential features [70]. Hence, we propose a multiscale spectral difference strategy in Fig. 4 and perform five different difference enhancements for four results of multiscale spatial enhancement. Finally, 20 enhanced results are obtained as follows:

$$\mathbf{P}_{\mathrm{ss}}^{mn} = \begin{cases} \mathbf{P}_{\mathrm{spa}}^{m} & n = 0 \\ \mathbf{P}_{\mathrm{spa}}^{m}[:,:,n:] - \mathbf{P}_{\mathrm{spa}}^{m}[:,:,:C-n] & n \neq 0 \end{cases} \tag{3}$$

where n is the scale of the spectral difference strategy chosen from $\{0, 1, 2, 4, 8\}$. $\mathbf{P}_{\mathrm{ss}}^{mn}$ denotes the results of spatial and spectral enhancement on the mth spatial scale and the nth spectral scale. C is the channel number of the original HSI. When $n = 0$, $\mathbf{P}_{\mathrm{ss}}^{mn}$ is the original spatial feature.

3) Comparison With Analogous Methods: Most of the prevailing data augmentation is based on computer Vision enhancement methods, for example, Ballester and Araujo [71] used the RandomResizedCrop strategy, and [72] applied random cropping and flipping. However, these methods do not consider the characteristics of hyperspectral images, such as the spectral continuity of HSIs [45] and multiscale characteristics of ground objects [51]. Hence, we propose the SSES to improve the discriminability of features. Specifically, we extract the spatial patch of four scales to improve the discriminability of ground objects of different sizes. Besides, the multiscale spectral difference strategy can rich spectral characteristics of the image, thereby enhancing the discriminability of spectral features.

D. Adaptive Decision Fusion

When training samples are few, deep learning models often suffer from overfitting [33]. Considering that if there is only a single prediction, it is difficult to detect the overfitting prediction part. We exploit the diverse spatial–spectral features

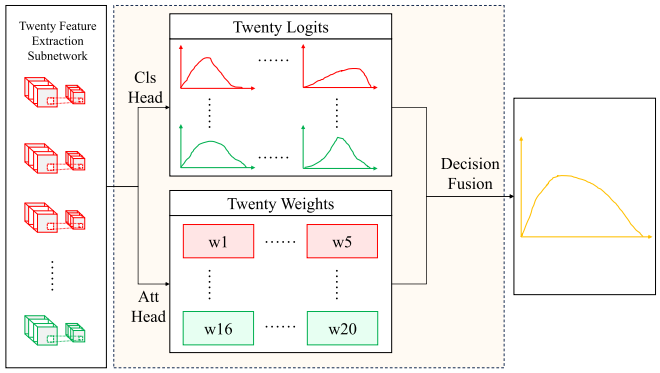


Fig. 5. Overview of ADF. Twenty spatial–spectral enhancement features are fed into the corresponding subnetwork to extract the patch feature $\mathbf{h}_{mn}$. Then, these patch features serve as the inputs of the classification head (cls head) and the attention head (att head) to obtain the logits and corresponding fusion weights. Finally, logits with different overfitting trends are weightedly fused using the weights estimated by the attention heads.

P_{ss}^{mn} generated by the SSES to derive multiple predictions. Because different predictions may have different trends of overfitting, integrating multiple predictions can help alleviate overfitting [65], [66]. Furthermore, we introduce an attention mechanism to estimate the fusion weight of each prediction. The proposed ADF module is shown in Fig. 5. Specifically, we first obtain multiple predictions and fusion weight using subnetworks as follows:

$$\mathbf{l}^{mn}, \mathrm{att}^{mn} = \mathrm{SN}^{mn}\left(\mathbf{P}_{\mathrm{ss}}^{mn}\right) \tag{4}$$

where $\mathbf{l}^{\mathbf{mn}}$, att^{mn}, and SN^{mn} denote the corresponding logits, fusion weight, and subnetwork of the $\mathbf{P}_{\mathrm{ss}}^{mn}$ feature, respectively. $\mathbf{P}_{\mathrm{ss}}^{mn}$ denotes the feature of spatial and spectral enhancement at the mth spatial scale and the nth spectral scale. Each subnetwork SN^{mn} consists of three components: an encoder Enc that extracts patch features $\mathbf{h}^{mn} = \mathrm{Enc}(\mathbf{P}^{mn})$, a classifier head Cls that generates logits $\mathbf{l}^{mn} = \mathrm{Cls}(\mathbf{h}^{mn})$, and an attention head Att that determines fusion weights $\mathrm{att}^{mn} = \mathrm{Att}(\mathbf{h}^{mn})$. The encoders in all subnetworks are implemented as CNNs with an integrated global average pooling layer. The classifier head consists of a multilayer perceptron (MLP) with K output nodes, where K represents the number of classes. The attention head employs an MLP with a sigmoid activation function.

Then, all logits are fused by the proposed adaptive fusion module as follows:

$$\mathbf{l}^{\mathrm{fus}} = \frac{1}{N^{mn}} \sum_{m}^{M} \sum_{n}^{N} \mathrm{att}^{mn} \mathbf{l}^{mn} \tag{5}$$

where $\mathbf{l}^{\mathrm{fus}}$, M, and N denote the fusion logits, the spatial scale set $\{1, 3, 9, 27\}$, and the spectral scale set $\{0, 1, 2, 4, 8\}$, respectively. $N^{mn} = N^{m} \times N^{n}$, where N^{m} and N^{n} denote the number of spatial scales and spectral scales, respectively.

E. Training and Inference

The training and inference procedure is presented in Algorithm 1. The training loss function $\mathcal{L}$ is given by the following equation:

$$\mathcal{L} = \mathcal{L}_{\mathrm{fus}} + \mathcal{L}_{\mathrm{sub}} \tag{6}$$

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Algorithm 1 Training and Test Procedures of SSEFN

```
// Training
Input: A 27 × 27 Patch P_{i,j} and training label Y_tr
Initialize: Initialize network parameters Θ with random values
for e = 1, 2, ..., E do
    // Spatial-Spectral Enhancement
    P_ss^mn ← {P_{i,j}} in Eq.(2 and 3),
    // Logits and Fusion Weights
    l^mn, att^mn ← {P_ss^mn} in Eq.(4),
    // Adaptive Decision Fusion
    l^fus ← {l^mn, att^mn} in Eq.(5),
    // Compute Loss
    L ← {l^mn, l^fus, Y_tr} in Eq.(6),
    // Update Parameters
    Update the parameters of SSEFN using Adam.
end
Output: Network parameters Θ
// Testing
Input: A 27 × 27 Patch P_te
1) Load trained network parameter Θ,
2) Enhance the test patch using SSES,
3) Obtain fusion logits using Eq.(4 and 5),
4) Predict the final class using Eq.(9).
Output: The class prediction ŷ
```

where $\mathcal{L}_{\text{fus}}$ and $\mathcal{L}_{\text{sub}}$ denote the loss of fusion logits and all logits from subnetworks, respectively. $\mathcal{L}_{\text{sub}}$ constrains the output of all subnetworks to predict labels as accurately as possible. The loss of subnetworks is given by the following equation:

$$\mathcal{L}_{\text{sub}} = \frac{1}{N^{mn}} \sum_{m}^{M} \sum_{n}^{N} \text{CrossEntropy}\left(\mathbf{l}^{\mathbf{mn}}, \mathbf{Y}_{\text{tr}}\right) \tag{7}$$

where $\mathbf{Y}_{\text{tr}}$ denotes the label of training data and CrossEntropy signifies cross-entropy loss. Besides, we demand that the final fused logits are predicted as accurately as possible. The fusion loss is defined as follows:

$$\mathcal{L}_{\text{fus}} = \text{CrossEntropy}\left(\mathbf{l}^{\text{fus}}, \mathbf{Y}_{\text{tr}}\right). \tag{8}$$

In inference phase, the final class prediction $\hat{y}$ can be obtained according to

$$\hat{y} = \arg\max\left(\text{softmax}\left(\mathbf{l}^{\text{fus}}\right)\right). \tag{9}$$

IV. Experiments

In this section, we first describe the experimental setup, including datasets, evaluation metrics, comparison methods, and implementation details. Then, we conduct a comprehensive comparison with state-of-the-art approaches from a qualitative and quantitative perspective. Finally, we perform detailed ablation studies to analyze the effect of the proposed modules.

A. Experimental Setting

Datasets: To fairly demonstrate the effectiveness of the proposed SSEFN, we choose four widely used and diverse hyperspectral datasets, that is, Indian Pines (IP), University of Pavia (UP), HanChuan, and Houston2013.

1) Indian Pines: The dataset, collected by AVIRIS [73] sensor on the IP agricultural test site in north-western Indiana in June 1992, contains 145 × 145 pixels with a spatial resolution of 20 m per pixel. After removing 20 water absorption and low SNR bands, the 200 bands with 10 249 labeled pixels are used for the analysis. Additionally, the dataset includes 16 classes.

2) University of Pavia: The dataset was captured by the reflective optics system imaging spectrometer (ROSIS) [74], over Pavia, Northern Italy. It is made up of 103 bands with a spatial size of 610 × 340. The UP dataset contains 42 776 labeled pixels of nine urban classes.

3) HanChuan: The dataset was acquired from 17:57 to 18:46 in June 17, 2016, in HanChuan, Hubei province, China, with a Headwall Nano-Hyperspec imaging sensor with 17-mm focal length equipped on a Leica Aibot X6 UAV V1 platform. The study area is a rural–urban fringe zone with buildings, water, and cultivated land, which contains seven crop species: strawberry, cowpea, soybean, sorghum, water spinach, watermelon, and greens. The image is composed of 274 bands from 400 to 1000 nm and has 1217 × 303 pixels. Sixteen categories are included in these data.

4) Houston2013: The dataset was acquired over the campus of the University of Houston and its surrounding areas, in Houston, TX, USA. This dataset was provided for the 2013 IEEE GRSS data fusion contest and contains 349 × 1905 pixels with 144 spectral channels. In this dataset, 15 main categories are investigated for the land-cover classification task.

a) Evaluation metric: In the experiments, all metrics are evaluated on the test set. In experiments, three evaluation metrics widely used in the classification of HSIs are adopted, including the overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa). To eliminate the deviation caused by randomly selecting training samples, all experiments are repeated ten times, and the mean and standard deviation values of each evaluation metric are reported. The higher the mean value and the smaller the standard deviation of these metrics are, the better the performance of the model is.

b) Comparison methods: We compare relative hyperspectral image classification methods.

1) *SSFTT[TGRS22] [54]:* The model combines the hierarchical CNN module and the transformer structure to capture spectral–spatial features and high-level semantic features.
2) *GAHT[TGRS22] [75]:* The model proposes a grouped pixel embedding (GPE) to emphasize local relationships in HSI spectral channels, yielding a global–local fashion from a spatial–spectral context for HSI classification.
3) *CVSSN[TCSVT22] [27]:* The model exploits the relationship between center and neighborhoods for HSI classification by adopting a central vector-oriented self-similarity network.
4) *CLOLN[TGRS24] [76]:* The model is a channel-layer-oriented lightweight network and can alleviate the number of parameters and computational complexity associated with CNNs.
5) *SQSFormer[TGRS24] [28]:* The model achieves adaptive spectral–spatial information fusion using the

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TABLE I
QUANTITATIVE RESULT (ACC% ± STD%) OF IP. BEST IN **BOLD** AND SECOND WITH UNDERLINE

Class	CVSSN TCSVT2022	SSFTT TGRS2022	GAHT TGRS2022	CLOLN TGRS2024	MambaHSI TGRS2024	SQSFormer TGRS2024	Res2ETF TGRS2024	ReSC-Net TGRS2024	SSEFN Ours
Alfalfa	55.30±20.02	49.62±27.08	43.62±16.97	50.59±26.21	**96.67±2.42**	77.40±15.62	58.69±23.41	73.14±19.87	74.18±18.79
Corn-notill	57.40±12.49	54.65±11.25	51.78±12.49	58.99±16.31	45.35±14.03	62.24±14.61	70.69±9.71	72.88±10.44	**75.00±12.73**
Corn-mintill	56.55±11.69	46.08±11.99	34.55±11.11	48.52±16.87	38.41±16.12	41.13±4.11	**78.23±11.81**	65.61±12.60	70.71±12.00
Corn	45.77±10.85	38.49±11.38	43.54±11.96	62.45±12.37	**80.62±10.16**	32.59±12.07	52.86±11.38	48.30±12.95	51.97±17.55
Grass-pasture	74.05±13.62	83.32±9.65	58.89±11.38	83.13±12.80	54.69±22.72	82.22±7.31	84.14±12.92	84.70±19.53	**91.89±8.59**
Grass-trees	89.85±2.52	93.11±2.61	78.53±10.12	91.35±6.88	72.81±14.89	93.47±6.63	92.95±4.82	**93.67±3.38**	**96.46±3.29**
Grass-pasture-mowed	26.70±12.06	35.49±25.44	32.69±9.51	31.79±13.28	**100.00±0.00**	27.75±8.49	28.01±8.70	26.35±3.38	34.62±17.34
Hay-windrowed	96.59±4.30	97.85±1.44	97.20±2.26	96.61±6.34	97.07±3.02	92.80±7.22	**98.41±2.02**	97.44±3.37	98.40±2.58
Oats	14.19±6.25	26.75±25.78	9.35±4.60	14.74±7.30	**100.00±0.00**	16.64±5.87	21.33±23.69	9.59±2.23	34.14±13.88
Soybean-notill	47.19±10.28	51.44±9.60	44.39±11.43	52.4±13.73	62.23±15.04	62.27±9.81	66.94±10.06	**77.24±15.32**	71.85±12.31
Soybean-mintill	72.35±8.40	69.79±11.62	67.10±5.47	81.76±11.82	47.77±13.37	81.45±7.80	88.47±8.51	**90.99±6.20**	83.79±6.60
Soybean-clean	44.93±16.97	39.93±14.02	38.59±10.00	40.44±12.50	46.62±7.67	43.37±11.47	55.18±14.82	56.08±14.04	**57.48±15.51**
Wheat	80.64±11.27	93.44±5.63	66.59±18.18	77.11±13.87	98.00±2.14	83.51±7.85	81.21±8.86	77.57±9.57	**94.45±3.37**
Woods	90.94±4.99	90.62±4.85	86.20±4.26	92.31±6.78	91.23±4.33	91.50±4.45	94.65±3.07	**97.55±1.89**	96.60±2.44
Buildings-Grass-Trees-Drives	71.24±13.07	63.97±10.54	66.00±11.13	71.81±13.24	38.11±19.31	77.78±15.63	68.87±11.50	71.92±14.12	**81.92±9.67**
Stone-Steel-Towers	66.82±11.56	66.64±18.24	43.16±11.15	41.23±18.55	**99.64±0.55**	66.11±15.83	54.17±8.42	50.14±11.92	74.19±12.60
OA(%)	65.28±3.48	63.37±4.76	58.04±2.45	64.07±4.91	59.89±4.65	68.87±3.17	76.45±4.40	76.28±3.78	**78.40±2.68**
AA(%)	61.91±3.36	62.57±3.94	53.89±2.90	62.20±4.15	73.08±2.84	64.82±2.47	68.66±3.13	68.32±3.18	**74.23±2.41**
$\kappa \times 100$	61.01±3.68	58.97±4.87	52.96±2.62	60.04±5.12	70.80±3.24	64.93±3.54	73.81±4.86	73.38±4.14	**75.57±2.94**

TABLE II
QUANTITATIVE RESULT (ACC% ± STD%) OF UP. BEST IN **BOLD** AND SECOND WITH UNDERLINE

Class	CVSSN TCSVT2022	SSFTT TGRS2022	GAHT TGRS2022	CLOLN TGRS2024	MambaHSI TGRS2024	SQSFormer TGRS2024	Res2ETF TGRS2024	ReSC-Net TGRS2024	SSEFN Ours
Asphalt	96.15±3.41	86.61±5.87	92.74±4.62	94.79±3.85	72.91±8.79	90.90±11.73	93.94±2.49	94.70±3.72	**97.57±3.03**
Meadows	92.11±4.20	93.16±3.89	90.15±3.42	95.94±3.39	73.17±18.98	92.21±3.10	94.97±2.47	**96.84±2.85**	96.14±2.64
Gravel	60.19±16.11	45.11±12.75	39.52±9.25	64.30±18.67	62.17±13.09	52.28±14.48	66.80±17.83	**72.11±17.31**	64.59±11.55
Trees	**83.51±20.30**	78.54±13.72	71.23±19.96	78.18±18.72	77.32±9.55	77.29±17.06	70.34±19.40	67.14±20.70	76.40±20.12
Metal sheets	94.52±11.64	99.39±0.68	96.61±3.12	98.98±0.91	99.29±0.90	**99.67±0.29**	96.45±2.82	97.79±1.92	**99.70±0.52**
Bare soil	46.63±12.09	60.22±17.84	48.46±9.72	51.08±9.68	57.63±20.49	50.62±8.49	57.04±11.47	63.58±16.74	**68.73±18.24**
Bitumen	45.98±9.75	63.03±14.71	35.53±10.11	61.50±17.87	**86.64±7.14**	44.46±7.66	63.87±24.67	72.25±16.24	71.87±15.01
Bricks	73.01±7.90	57.76±11.92	67.48±14.36	72.13±6.49	68.66±11.97	71.41±7.36	**79.28±8.22**	78.93±4.40	74.80±5.61
Shadows	92.07±9.41	64.78±16.72	77.58±18.08	91.61±9.54	**98.42±0.99**	83.83±14.91	90.20±7.74	88.35±9.62	93.73±6.01
OA(%)	73.67±6.42	72.44±5.96	69.26±5.19	76.90±7.22	72.46±6.68	74.19±4.83	78.57±3.84	80.49±5.53	**82.71±6.31**
AA(%)	76.02±3.86	72.07±4.04	68.81±5.39	78.72±4.66	77.36±2.83	73.63±4.10	79.21±3.84	81.30±3.92	**82.62±3.35**
$\kappa \times 100$	67.10±7.25	65.70±6.53	61.41±5.75	71.32±7.93	75.50±6.96	67.53±5.50	73.08±4.51	75.68±6.15	**78.19±7.10**

Spectral Query Spatial paradigm, to reduce the introduction of irrelevant information and effectively improve classification performance.

6) *MambaHSI[TGRS24] [29]:* The model is an image-level HSI classification model that introduces Mamba to model long-range interaction of the whole image and extract spatial–spectral information.
7) *Res2ETF [TGRS24] [15]:* This model introduces a no-learned ETF classifier as well as an attention-enhanced resblock-in-resblock to improve the classification performance of limited samples.
8) *ReSC-Net [TGRS24] [18]:* The model introduces a new sample augmentation technique to address the challenge of limited samples and proposes a novel simple spatial–spectral attention-based model for hyperspectral image classification with only ten samples.

c) Implement details: The proposed SSEFN framework is implemented on the PyTorch platform and all runs are performed on the same hardware: NVIDIA GeForce RTX 3090 GPU, ×64 Inter Xeon Gold 6226R CPU, and 256 GB RAM. All experiments are carried out repeatedly ten times, where randomly select five samples from the labeled dataset for both the training and test sets, respectively, and then the remainder of the test set. We employ the Adam optimizer [77] with a learning rate initialized with 0.001. The size of the patch is set to 27.

B. Comparison With State-of-the-Arts

To fairly evaluate the performance and efficiency of the proposed model, we comprehensively compare our method with several SOTA HSI classification methods from a quantitative and qualitative perspective. Quantitative and qualitative evaluations are reported in Tables I–IV and Figs. 6–9, respectively. From the quantitative results, we find that Res2ETF and ReSC-Net as few labeled methods and their accuracies are higher than other methods. Taking the IP dataset as an example, Res2ETF outperforms the most competitive SQSFormer by improvements 9.53%, 9.41%, and 10.64% in OA, AA, and Kappa, respectively. This may be because other methods have overfitting problems caused by not considering scenarios with few training samples. More importantly, our proposed SSEFN model achieves the most outstanding performance compared to the others on all four datasets. This proves that our method has good adaptability to diverse environments.

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TABLE III
QUANTITATIVE RESULT (ACC% ± STD%) OF THE HANCHUAN DATASET. BEST IN **BOLD** AND SECOND WITH UNDERLINE

Class	CVSSN TCSVT2022	SSFTT TGRS2022	GAHT TGRS2022	CLOLN TGRS2024	MambaHSI TGRS2024	SQSFormer TGRS2024	Res2ETF TGRS2024	ReSC-Net TGRS2024	SSEFN Ours
Strawberry	79.22±4.98	82.69±5.95	86.00±8.75	82.72±12.34	69.64±9.19	91.55±3.41	90.57±2.19	**92.80±1.99**	92.12±4.44
Cowpea	81.42±8.58	84.11±10.92	70.39±6.95	80.53±9.74	46.56±7.51	76.90±8.19	82.35±9.51	76.04±5.47	**93.33±2.76**
Soybean	37.67±6.54	52.22±9.58	40.61±8.47	49.54±6.58	**76.04±13.65**	48.35±9.92	65.00±11.51	66.49±13.27	70.88±8.75
Sorghum	59.60±19.65	68.89±14.30	41.90±8.07	70.31±18.18	**95.15±1.90**	67.74±19.83	86.45±9.44	83.65±12.35	80.46±10.25
Water spinach	23.06±14.71	19.77±4.17	23.70±9.22	18.00±12.03	**95.55±3.44**	38.65±12.31	36.92±11.45	19.44±7.35	25.88±7.40
Watermelon	13.29±4.86	22.12±5.81	14.94±3.73	21.23±16.09	37.61±9.53	27.46±8.86	28.11±8.08	29.58±12.47	**38.28±6.75**
Greens	51.83±16.53	41.91±11.88	42.82±6.81	40.90±18.16	**93.74±4.90**	51.56±17.78	49.36±18.33	49.05±11.26	49.94±8.67
Trees	60.70±11.96	60.90±14.77	54.18±16.35	62.77±13.93	37.27±8.23	62.42±16.17	68.11±11.24	67.94±13.92	**87.55±6.77**
Grass	44.26±14.50	51.71±11.57	42.32±11.54	49.97±19.13	33.26±7.09	57.91±13.48	58.73±17.71	59.31±11.02	**66.11±8.26**
Red roof	72.84±27.36	65.78±15.38	69.44±13.54	68.50±25.97	68.04±16.21	66.88±19.36	84.63±15.72	**85.95±18.56**	83.78±14.40
Gray roof	**76.83±21.09**	51.36±19.50	61.17±14.49	59.97±22.55	75.75±16.69	62.35±19.06	61.74±20.07	56.27±18.14	53.37±12.00
Plastic	31.68±19.83	28.02±9.63	17.73±1.42	23.62±10.40	**55.56±12.61**	40.61±12.04	28.52±11.84	21.26±11.08	45.89±17.02
Bare soil	30.62±16.34	29.92±10.14	32.89±7.10	35.01±12.01	42.51±9.34	42.07±9.79	43.49±13.04	49.27±12.94	**56.33±14.74**
Road	59.39±14.31	61.53±19.31	66.68±11.89	62.33±17.49	53.22±8.65	**75.93±13.32**	70.42±17.63	70.37±16.88	69.45±11.92
Bright object	56.03±16.38	28.75±11.51	27.14±11.76	48.81±18.29	**76.89±11.95**	40.90±16.71	35.14±15.01	35.88±20.92	44.93±24.06
Water	83.36±12.45	96.02±2.91	96.20±3.64	93.52±8.92	76.92±14.20	95.56±5.04	97.63±5.08	**98.57±3.49**	97.11±4.20
OA(%)	60.63±4.31	63.78±3.94	65.10±2.97	63.10±5.15	65.06±4.32	71.51±4.40	72.07±5.05	71.34±5.15	**76.33±3.24**
AA(%)	53.86±3.27	52.86±2.45	49.26±2.19	54.23±4.28	64.61±2.70	59.18±3.98	61.70±3.76	60.12±3.36	**65.96±3.03**
$\kappa \times 100$	53.97±4.50	58.52±4.25	59.66±3.51	57.55±5.61	64.40±3.63	67.05±4.83	67.93±5.43	67.14±5.65	**72.64±3.69**

TABLE IV
QUANTITATIVE RESULT (ACC% ± STD%) OF THE HOUSTON2013 DATASET. BEST IN **BOLD** AND SECOND WITH UNDERLINE

Class	CVSSN TCSVT2022	SSFTT TGRS2022	GAHT TGRS2022	CLOLN TGRS2024	MambaHSI TGRS2024	SQSFormer TGRS2024	Res2ETF TGRS2024	ReSC-Net TGRS2024	SSEFN Ours
Healthy Grass	76.82±15.23	84.51±12.38	84.69±8.99	70.80±9.54	**89.12±6.73**	76.71±12.73	86.1±5.76	84.99±7.37	83.03±9.13
Stressed Grass	85.05±18.48	86.06±11.08	81.48±6.68	86.42±13.58	84.64±8.68	75.25±8.17	**90.04±8.60**	86.92±9.58	89.65±8.60
Synthetic Grass	72.73±18.18	95.57±6.91	86.03±10.20	73.98±20.59	93.71±13.77	98.66±1.99	**99.35±1.22**	92.85±16.30	99.28±2.22
Tree	82.44±13.24	86.03±7.20	83.22±9.20	76.20±11.31	92.11±4.06	82.65±11.46	87.45±8.43	90.36±6.71	**91.39±7.41**
Soil	92.34±4.36	90.19±7.16	91.31±4.71	85.83±13.31	**98.09±2.50**	91.09±5.44	87.91±8.72	90.85±7.54	88.60±6.06
Water	70.90±17.30	94.53±10.88	60.89±10.93	73.61±22.97	79.40±10.34	69.10±22.13	**96.86±5.37**	92.92±11.30	94.48±11.63
Residential	80.53±10.99	77.23±10.76	76.90±7.84	75.38±13.60	68.74±13.26	80.19±5.58	76.07±11.80	76.45±8.69	**82.87±8.59**
Commercial	79.85±18.20	73.67±20.75	77.60±17.01	76.71±22.62	35.30±8.67	63.78±10.38	79.62±17.19	**84.90±15.10**	78.38±12.96
Road	**79.66±13.27**	69.30±17.00	**80.34±7.57**	77.92±14.39	67.21±12.09	68.72±10.76	75.92±11.24	71.60±10.90	72.70±13.08
Highway	48.92±12.17	60.91±10.94	55.41±10.96	49.76±18.50	53.30±16.20	56.61±12.62	**66.77±12.51**	59.13±9.40	65.19±13.44
Railway	63.02±15.69	63.93±14.76	63.79±12.47	57.79±16.04	58.66±10.83	**73.67±11.11**	71.40±14.67	64.13±21.40	70.54±13.10
Parking Lot 1	54.57±7.40	58.66±15.03	56.06±7.34	52.78±14.43	60.88±14.74	61.94±7.77	67.72±10.86	**69.47±10.30**	69.25±11.82
Parking Lot 2	71.03±19.42	83.67±8.08	67.88±18.78	70.65±8.45	82.09±8.15	**88.09±14.10**	59.01±15.36	45.78±9.37	73.15±16.18
Tennis Court	89.26±11.13	98.07±3.15	93.06±4.72	79.89±9.76	**99.90±0.22**	96.70±4.37	93.66±10.24	94.04±9.10	97.78±3.42
Running Track	90.11±7.23	93.64±4.42	92.68±5.08	92.84±5.32	91.86±12.67	**95.06±2.39**	90.89±3.46	94.27±2.08	93.04±4.03
OA(%)	71.46±4.36	75.25±3.65	74.64±2.62	68.00±2.85	74.17±3.85	74.30±1.16	78.20±2.11	76.32±2.48	**79.67±1.75**
AA(%)	75.82±4.70	81.07±2.43	76.76±2.13	73.37±1.83	77.00±3.18	78.55±1.39	81.92±1.44	79.91±2.44	**83.29±2.08**
$\kappa \times 100$	69.20±4.69	73.24±3.96	72.63±2.82	65.48±3.06	73.43±3.95	72.21±1.25	76.45±2.28	74.44±2.65	**78.02±1.89**

Taking the HanChuan dataset as an example, the proposed SSEFN model outperforms the most competitive counterpart Res2ETF by 4.26%, 4.26%, and 4.71% in OA, AA, and Kappa, respectively.

From qualitative evaluation, the classification maps produced by the corresponding methods in the IP, UP, HanChuan, and Houston2013 datasets are shown in Figs. 6–9, respectively. Totally, in comparison with other methods, the proposed SSEFN model has less misclassification, producing a high-quality classification map with smooth objects and well-maintained boundaries. Taking the HanChuan dataset as an example, all counterparts contain a lot of noise areas, misclassification areas, and blurring boundaries. In contrast, the proposed SSEFN model exhibits better object continuity and a clearer boundary.

In summary, compared to the state-of-the-arts, our proposed SSEFN can achieve the best classification performance in all quantitative metrics and visualization. This demonstrates the superiority of the proposed SSEFN.

C. Ablation Study

To further explore and validate the contributions of different modules of the proposed SSEFN model, ablation experiments are conducted on four difficult datasets, IP, UP, HanChuan, and Houston2013, to analyze the effect of each component, including SpaES, SpeES, and ADF modules. As recorded in Table V, we can observe that: 1) both the proposed SpaES and SpeES modules are significantly better than simply repeating fixed spatial scales and spectral scales, which indicates that the strategy of generating multiple results and integrating them through repeated networks does not work well. This demonstrates the effectiveness of introducing multiscale spatial and multiscale differential spectral information; 2) using SpaES and SpeES together is better than using either SpaES or SpeES alone. Taking the HanChuan dataset as an example, the performance of SpaES and SpeES improves 9.99% and 28.19% in OA over SpaES and SpeES, respectively. This proves that both SpaES and SpeES are critical and complementary for enhancing the discriminability of

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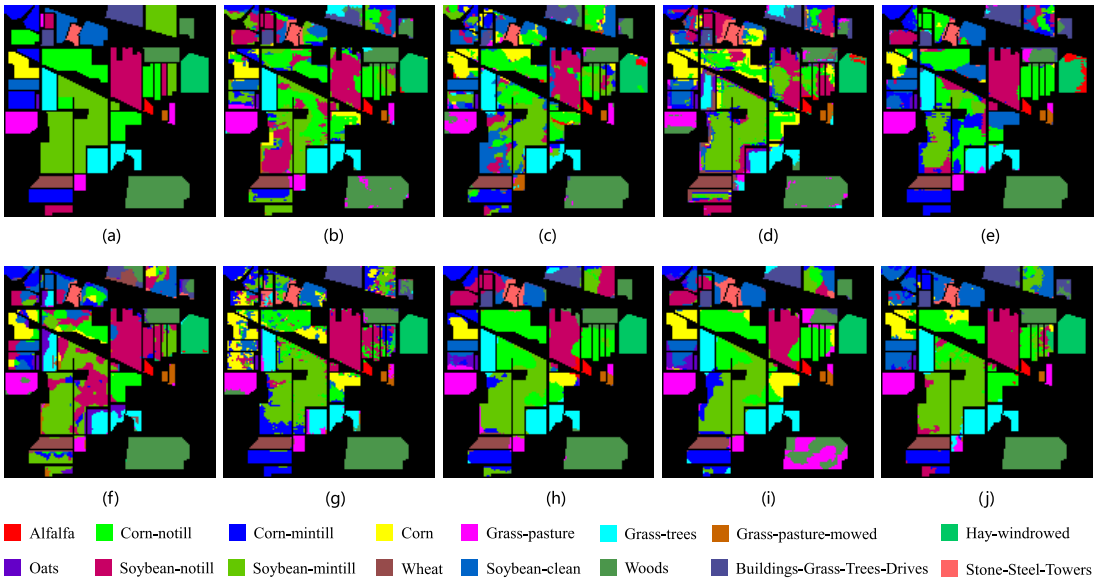


Fig. 6. Qualitative visualization of the classification map for the IP dataset. (a) Ground-truth map. (b) CVSSN. (c) SSFTT. (d) GAHT. (e) CLOLN. (f) MambaHSI. (g) SQSFormer. (h) Res2ETF. (i) ReSC-Net. (j) Proposed SSEFN.

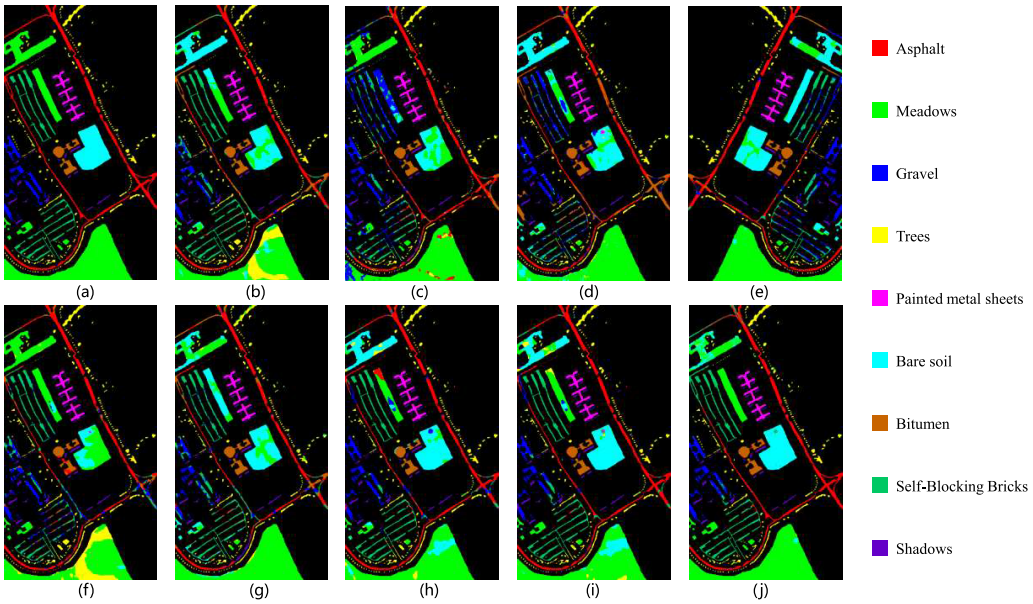


Fig. 7. Qualitative visualization of the classification map for the UP dataset. (a) Ground-truth map. (b) CVSSN. (c) SSFTT. (d) GAHT. (e) CLOLN. (f) MambaHSI. (g) SQSFormer. (h) Res2ETF. (i) ReSC-Net. (j) Proposed SSEFN.

TABLE V

ABLATION STUDY OF KEY COMPONENTS OF OUR METHOD. SpaES, SpeES, ADF, AND FEATURE DENOTE THE SPATIAL BLOCK, SPECTRAL BLOCK, ADAPT DECISION FUSION MODULE, AND FEATURE FUSION, RESPECTIVELY. RepeatSpa SETS ALL FOUR SPATIAL SCALE AS 9 TO VALIDATE THE EFFECT OF MULTISCALE SPATIAL ENHANCEMENT. REPEATSPE SETS ALL FIVE SPECTRAL SCALE AS 1 TO ANALYZE THE EFFECT OF MULTISCALE SPECTRAL ENHANCEMENT

SpaES	SpeES	ADF	Feature	RepeatSpa	RepeatSpe	IP	UP	HanChuan	Houston2013
		✓		✓		68.09±5.59	76.11±6.56	66.10±3.70	70.10±4.27
		✓			✓	36.72±2.46	50.11±4.42	23.00±3.84	47.44±2.10
✓		✓				72.56±4.33	79.63±2.59	66.34±4.84	74.01±3.62
	✓	✓				48.08±3.00	61.69±4.35	48.14±2.85	65.74±1.55
✓	✓		✓			66.22±4.16	76.79±6.38	60.00±4.79	67.04±5.88
✓	✓	✓				**78.40±2.68**	**82.71±6.31**	**76.33±3.24**	**79.67±1.75**

spatial–spectral features; and 3) the results of ADF are higher than those of feature fusion, which shows that decision-level fusion is more effective in reducing over-fitting problems in few labeled samples. Taking Houston2013, for instance, the

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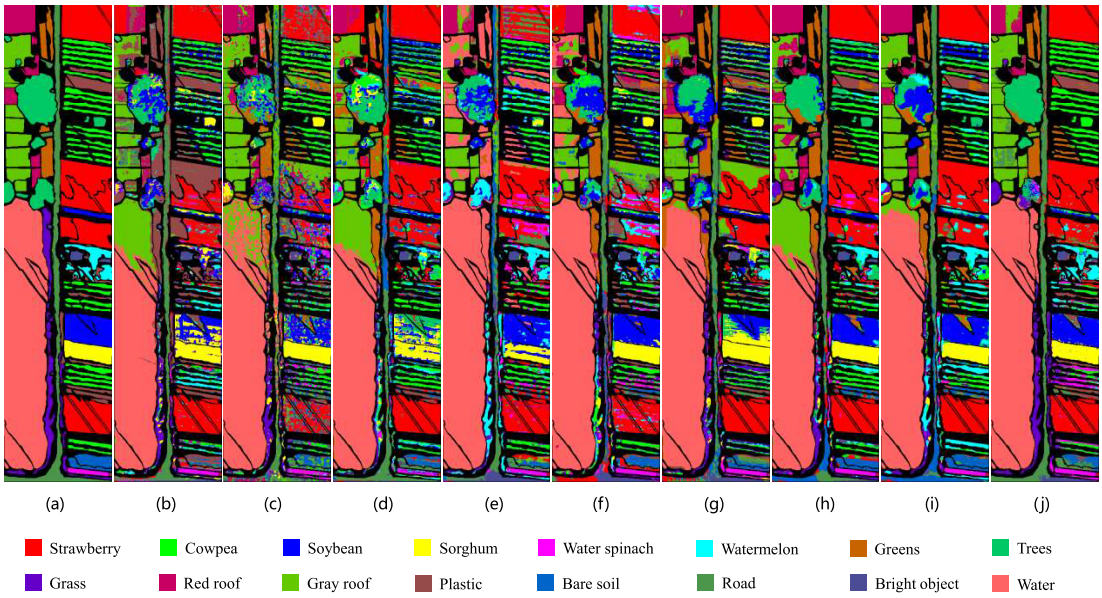


Fig. 8. Qualitative visualization of the classification map for the HanChuan dataset. (a) Ground-truth map. (b) CVSSN. (c) SSFTT. (d) GAHT. (e) CLOLN. (f) MambaHSI. (g) SQSFormer. (h) Res2ETF. (i) ReSC-Net. (j) Proposed SSEFN.

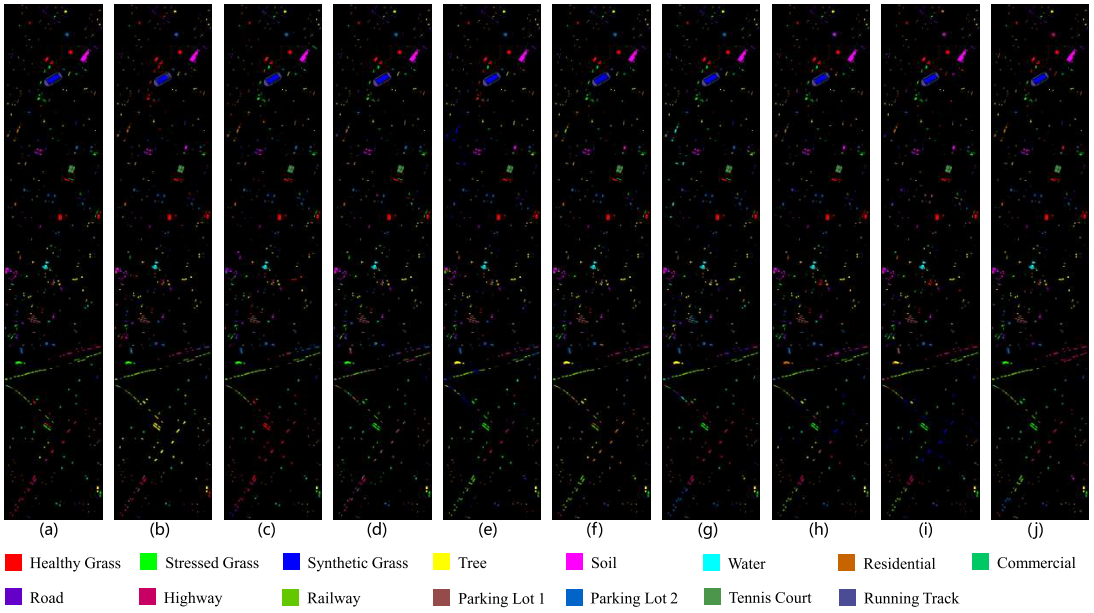


Fig. 9. Qualitative visualization of the classification map for the Houston2013 dataset. (a) Ground-truth map. (b) CVSSN. (c) SSFTT. (d) GAHT. (e) CLOLN. (f) MambaHSI. (g) SQSFormer. (h) Res2ETF. (i) ReSC-Net. (j) Proposed SSEFN.

performance of ADF improves 10.71% in OA over feature fusion.

To analyze the effect of ADF in alleviating overfitting, we further compared the overfitting trends of all subnetworks and fused logits. From the results in Fig. 10, we observe that: 1) the prediction accuracy of different subnetworks on the same category varies greatly, which reflects the different overfitting tendencies of different subnetworks; and 2) compared with the subnetwork logits, the logits fused by ADF are more stable in different categories and achieve the highest OA. This proves the effectiveness of the proposed ADF module in alleviating overfitting.

D. Robustness Analysis for Noise

To evaluate the robustness of noise models, we followed the setting of [78] and added Gaussian noise with zero mean and different standard deviations to the hyperspectral image. From the results in Fig. 11, we can observe that: 1) as the standard deviation of the noise increases from 0 to 0.5, the performance of all models degrades. This demonstrates that noise can cause demonstrates; and 2) compared to the most competitive ReSC-Net [18] method, where the OA drops from 80.49% to 62.41%, the performance of our method only decreases from 82.71% to 77.66% when the noise ranges from 0 to 0.5. This demonstrates the robustness of the proposed SSEFN to noise.

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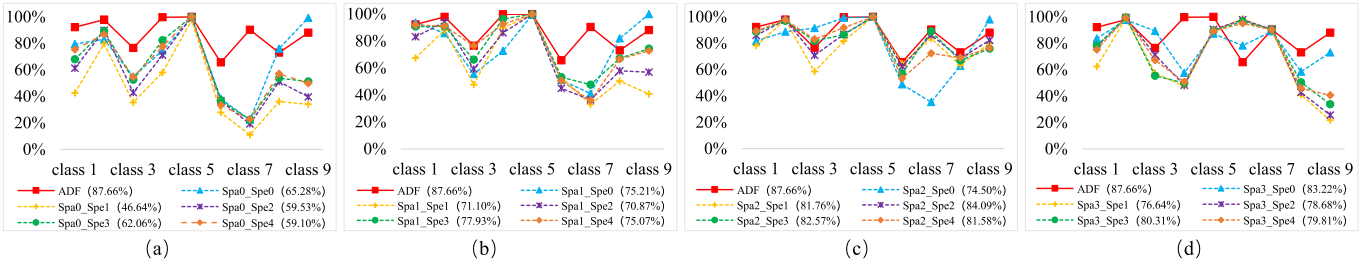


Fig. 10. Overfitting trends of all subnetworks and the fused module. To better visualize the difference between the predictions of individual subnetworks and the fused predictions, the outputs from 20 subnetworks are grouped into four subfigures (a)–(d) based on the spatial enhancement scales, and compared with the ADF fusion results. The numbers in parentheses in the legend indicate the OA of each branch across all categories.

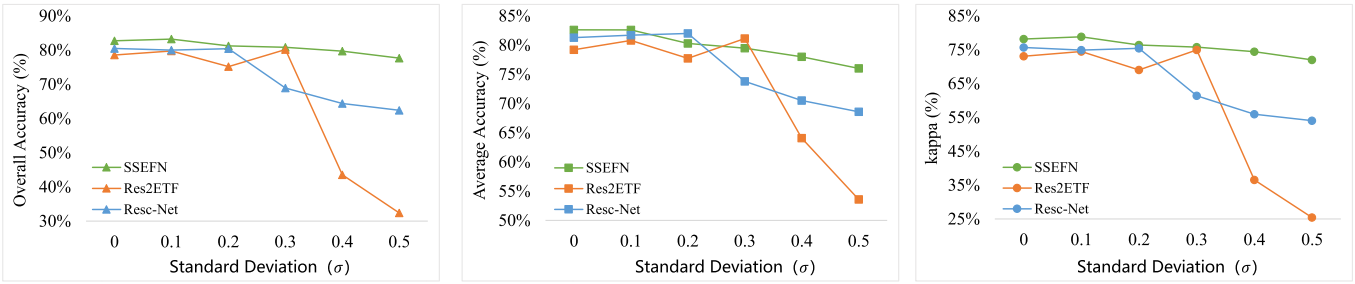


Fig. 11. Robust analysis for different levels of noise.

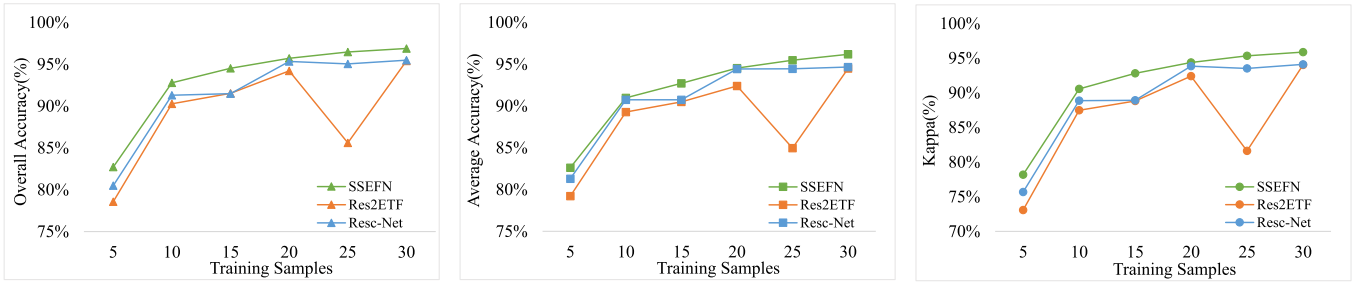


Fig. 12. Effect of sample size on UP.

E. Effect of Sample Size

We conducted experiments on varying sample sizes to evaluate the effect of sample size on model performance. From the results in Fig. 12, we can observe that as the number of samples increases from 5 to 30, the performance of the model improves rapidly, which demonstrates that the model can effectively utilize the new samples to improve performance. Specifically, the OA, AA, and kappa of each sample size in our proposed SSEFN model are higher than the few labeled models (Res2ETF [15] and Resc-Net [18]).

F. Performance on Cross-Dataset HSI Classification

To evaluate the generalizability of the proposed method, we conducted cross-dataset experiments on the cross-scene Pavia dataset [79]. The Pavia dataset includes the UP and the Pavia Center (PC). The PC has 1096 × 715 pixels and 102 bands. The UP has 103 spectral bands, 610 × 340 pixels, and 1.3-m spatial resolution, where the last band was removed to ensure the same number of spectral bands as PC. They all have the same seven classes.

Specifically, we train the model on the PC dataset and then test it on the PC and UP datasets, respectively, to evaluate the impact of cross-datasets on model performance. From the results in Table VI, we find that: 1) all models suffer from performance degradation in the cross-dataset setting. For example, when both the test set and the training set are PC datasets, Res2ETF can achieve 82.65% OA, but when the test set is inconsistent with the training set, the OA performance drops sharply to 38.69%; and 2) compared to the most competitive few-labeled HIS classification methods Res2ETF and ReSC-Net, the proposed SSEFN is more robust to the cross-dataset. This proves the superiority of the proposed SSEFN method.

Considering that the cross-dataset setting is very different from the few-shot setting studied in this article, it is understandable that the performance of the few labeled HSI classification models degrades in the cross-dataset setting. In the future, we will further study the hyperspectral domain generalization method for cross-datasets to improve the domain generalization of the model.

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TABLE VI

RESULTS OF CROSS-DATASET HSI CLASSIFICATION ON THE CROSS-SCENE PAVIA DATASET. WHEN THE TRAINING SET AND TEST SET ARE DIFFERENT, IT REPRESENTS A CROSS-DATASET SETTING, AND VICE VERSA

Training	Testing	OA(%)			AA(%)			Kappa(%)		
		Res2ETF TGRS2024	Resc-Net TGRS2024	SSEFN Ours	Res2ETF TGRS2024	Resc-Net TGRS2024	SSEFN Ours	Res2ETF TGRS2024	Resc-Net TGRS2024	SSEFN Ours
PC	PC	82.65±9.50	90.27±1.99	91.66±1.75	84.38±4.64	88.84±2.47	90.72±1.97	79.36±11.01	88.30±2.38	89.97±2.09
PC	UP	38.69±10.71	50.55±6.77	52.96±5.20	52.06±9.83	51.58±5.29	50.11±2.28	28.62±10.22	39.97±5.78	44.02±4.91

V. CONCLUSION

In this article, we present a spatial–spectral enhancement and fusion network for HSI classification, termed SSEFN, which has a strong feature extraction capability and is robust for overfitting when training samples are limited. The spatial and spectral enhancement strategy introduces the multiscale priors for ground objects and enriches the spectral features by adding spectral differential features. This improves the discriminability of the features, and the model can learn more easily with fewer samples. The ADF module fuses the decisions of all enhancement features. Since each decision may have different trends in overfitting, the integration of multiple decisions is beneficial in reducing overfitting. From the extensive experimental results on multiple datasets, we mainly conclude that: 1) introducing expert knowledge through enhancement strategy, such as multiscale prior of ground objects, is helpful to improve the performance when training data are few; and 2) decision fusion is beneficial for alleviating overfitting. In the future, we intend to extend the proposed idea to weakly supervised hyperspectral image classification.

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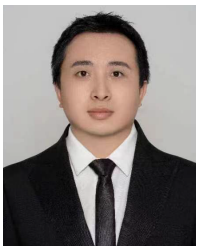
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